OUTPUT:

Data loading: Placeholder data used. Ensure to replace with actual finance data.

--- Q1 (Category 5): LDA Model Training and Evaluation (TM A) ---

LDA Model Accuracy: 0.3033

LDA Confusion Matrix:

```
[[22 50 28]
 [30 46 25]
 [24 52 23]]
```

LDA Feature Coefficients (First Discriminant, Standardized Data):

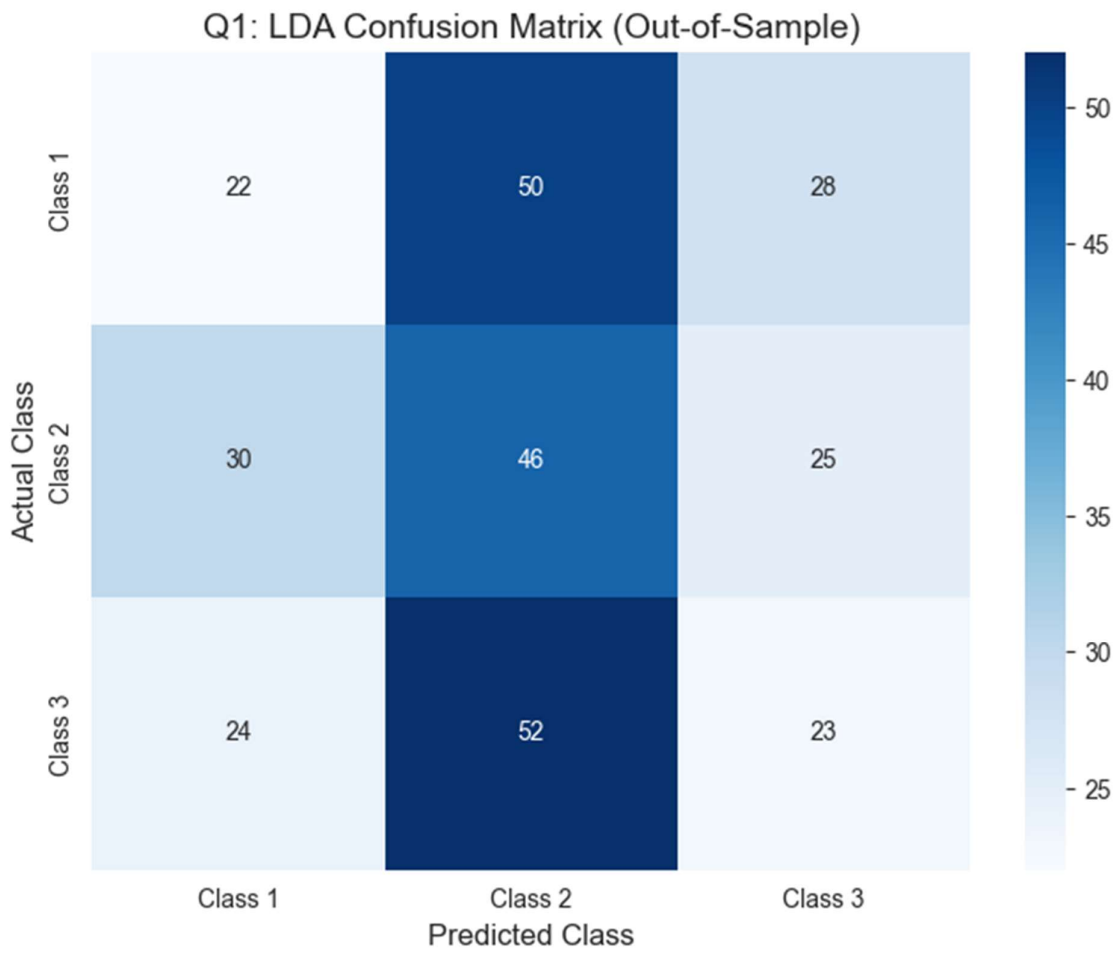
RSI_14: 0.0446

MACD_Signal: -0.0084

Vol_Index: -0.0561

Momentum: -0.0160

Market_Breadth: -0.0205



--- Q2 (Category 6): SVM Model Training and Evaluation (TM B) ---

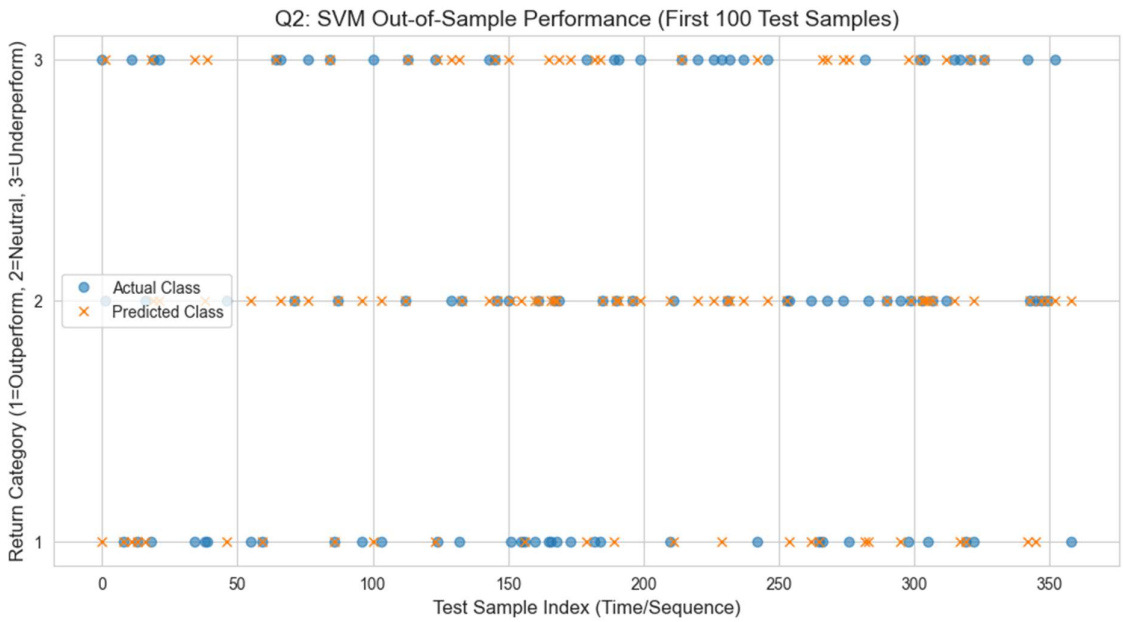
Optimal SVM Hyperparameters: {'C': 1, 'kernel': 'linear'}

Optimal SVM Model Accuracy: 0.3133

Optimal SVM Classification Report (F1-Score, etc.):

	precision	recall	f1-score	support
1	0.29	0.24	0.26	100
2	0.32	0.46	0.38	101
3	0.32	0.24	0.28	99
accuracy			0.31	300

macro avg	0.31	0.31	0.31	300
weighted avg	0.31	0.31	0.31	300

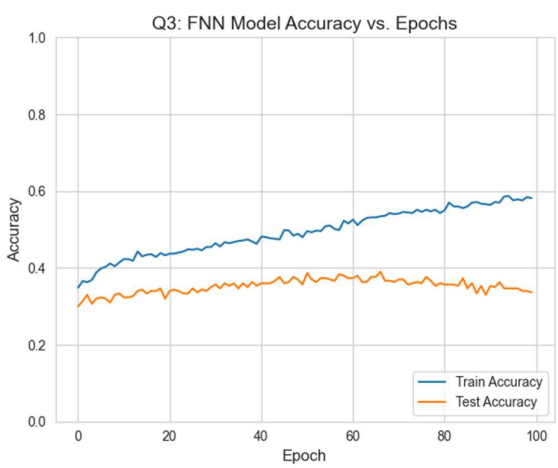


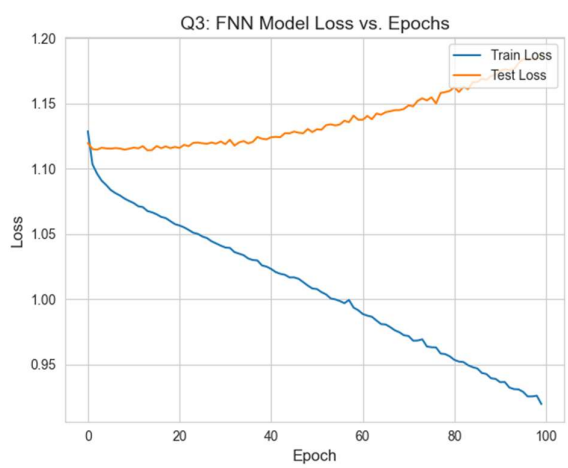
--- Q3 (Category 7): Neural Network Model Training and Evaluation (TM C) ---

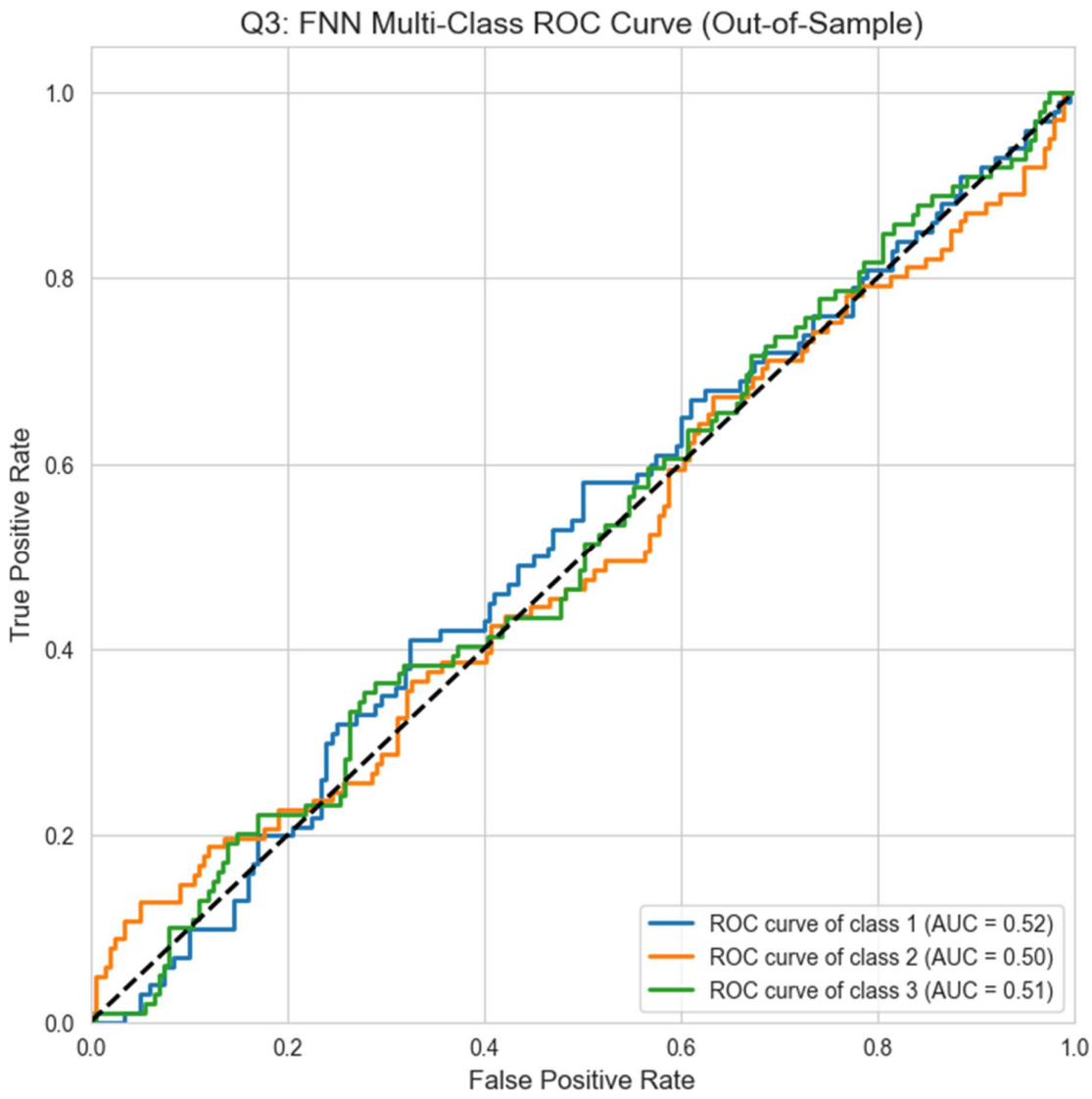
FNN Model Test Accuracy: 0.3367

FNN Model Test Loss: 1.1872

10/10 [==========] - 0s 1ms/step







--- All Quantitative Analysis computations complete. ---